CCNA 200-301 V2.0

29 Hypervisors, Virtual Machines and Containers

Part V — Wireless, Virtualisation and Operations

EXAM TOPICS

1.2

LECTURER

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LEARNING OUTCOMES

What you will be able to do

- **Distinguish** a type 1 from a type 2 hypervisor and say where each belongs.
- **Describe** what a virtual machine contains and how a virtual switch connects it to the physical network.
- **Compare** virtual machines with containers on isolation, start time, size and what they share.
- **Configure** the physical switch port a hypervisor host attaches to — which is the part of this topic that is actually your job.

ASSUMED KNOWLEDGE

Before we start

Chapter 8 and Chapter 10 — a hypervisor uplink is a trunk, and this chapter will not make sense without knowing what one is. Chapter 25 for port security, which is the wrong tool on such a port. Chapter 3 for MAC learning.

KEY TERMS

Vocabulary for this chapter

- hypervisor
- type 1
- type 2
- virtual machine
- guest OS
- vNIC
- virtual switch
- port group
- container
- image
- kernel
- orchestration
- east-west traffic
- live migration

Each is defined where it first matters — not here.

ON THE EXAM

Describe the role and function

This is what the blueprint actually asks for

Topic 1.2 is *describe the role and function of hypervisors, virtual machines, and containers*. No configuration of any hypervisor is required, and no vendor product is named.

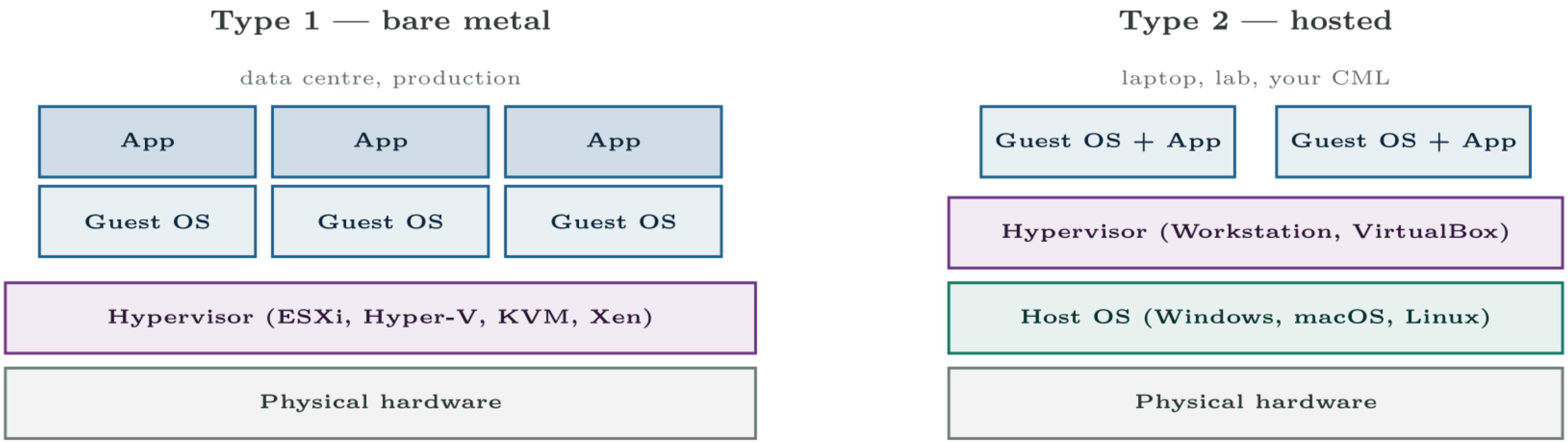
But it sits in domain 1, *Network Infrastructure and Connectivity*, and not by accident. A CCNA is not asked to administer a hypervisor; a CCNA is asked to **connect one correctly**, and to recognise the several things a virtualised server does to a network that a physical server never did. Section sec:vnetimpact is where those live, and it is the part of this chapter you will use at work next week.

DEFINITION

Hypervisor

Software that presents virtual hardware — CPU, memory, disk, network interface — to several guest operating systems at once, and schedules the real hardware between them. Also called a **virtual machine monitor**.

Type 1 and type 2



The difference is one layer. A type 1 hypervisor is the operating system; a type 2 hypervisor is an application running on one.

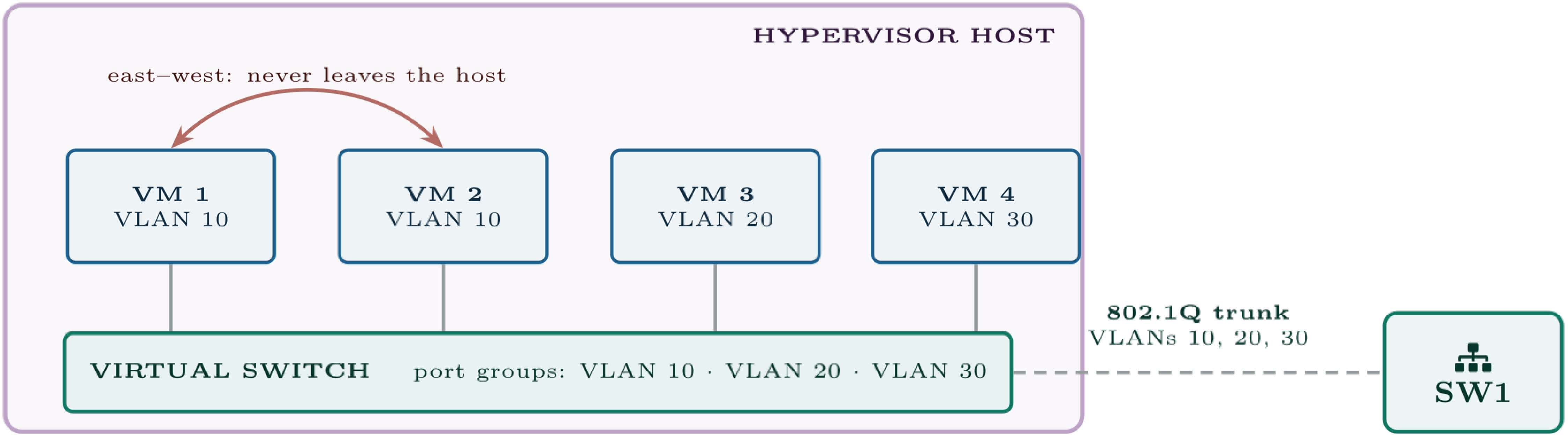
CONCEPT

You have almost certainly used both

The CML or GNS3 you run the labs in this book on is a type 2 hypervisor, or runs inside one. The cloud instances your university's systems run on are type 1.

If an exam question describes a hypervisor installed *onto* a server with no other operating system, it is type 1. If it describes one installed *into* Windows or macOS, it is type 2. That is the whole distinction.

The virtual switch



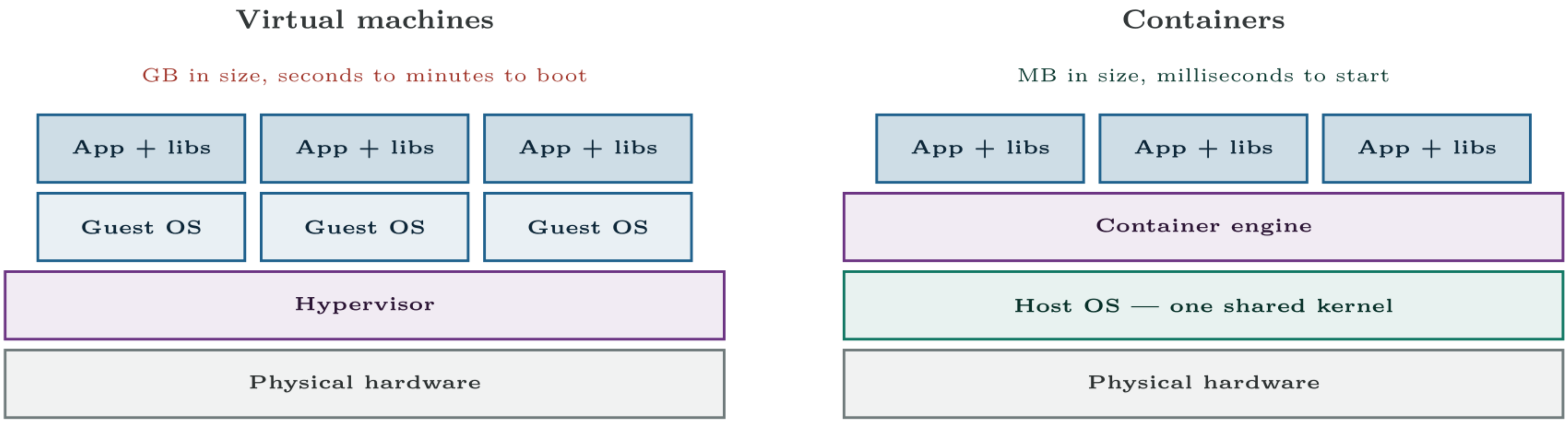
VM 1 talking to VM 2 never reaches SW1. Everything else does, over a trunk — a single physical port carrying three VLANs.

DEFINITION

Port group

A named policy on the virtual switch — a VLAN ID, security settings and traffic shaping — that VMs are attached to. It is the virtual equivalent of deciding which VLAN a switch port belongs to, and the VLAN ID configured on it must match one the physical switch actually trunks.

Containers



One layer removed. Containers share the host's kernel, which is why they are small and fast — and why the isolation between them is weaker.

CONCEPT

They are not competitors

The common real deployment is containers *inside* virtual machines: the VM provides hard isolation and the ability to move workloads between hosts, and the container provides fast, repeatable application packaging inside it. Every major cloud provider runs some version of this.

An exam question asking you to choose between them is asking about the properties in the table — kernel sharing, size, start time, isolation — not about which is better.

DEFINITION

Orchestration

Running one container by hand is easy. Running four hundred across thirty hosts, restarting the ones that fail, scaling with demand and routing traffic to whichever are healthy is not. An **orchestrator** — Kubernetes is the dominant one — does that scheduling, health-checking and service discovery.

For this exam, know what the word means and what problem it solves.

CONFIGURE

The physical switch port a hypervisor host attaches to

IOS · configuration mode

```
interface GigabitEthernet1/0/24
 description ESXi host 3 -- vmnic0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 switchport nonegotiate
!
! No port security: a hypervisor presents many MAC addresses and they
! change as machines start, stop and migrate.
!
! PortFast trunk, because the far side is a virtual switch that does
! not run spanning tree and will never send a BPDU.
 spanning-tree portfast trunk
!
! BPDU guard is still correct: nothing on this port should ever send
! a BPDU, so one arriving means something is wrong.
 spanning-tree bpduguard enable
```

WATCH OUT

The virtual switch does not run spanning tree

A vSwitch is not a learning switch in the ordinary sense: it does not forward traffic between its uplinks, so it cannot create a loop, and it does not participate in STP.

Two consequences. `spanning-tree portfast trunk` is appropriate on the physical port, because there is no risk of a loop and waiting 30 seconds on every host reboot is pure cost. And **BPDU guard should still be enabled** — precisely because a correctly configured vSwitch will never send a BPDU, one arriving means somebody has attached something else.

Common pitfalls

- **Access port on a hypervisor uplink.** Everything in one VLAN works, everything else is stranded, and the fault looks like a VM problem.
- **Port security on a hypervisor uplink.** Err-disables the port and takes down every VM on the host.
- **Reading migration MAC moves as a loop.** A MAC flap between two *access-layer uplinks* at the moment a VM migrated is expected behaviour.
- **Forgetting to add a VLAN to `allowed`** when the server team adds a port group. Their side is right, yours is not, and neither team sees both halves.
- **Expecting to capture east–west traffic** on the physical switch. It never gets there.

Common pitfalls (continued)

- **Assuming containers are just small VMs.** The shared kernel is the whole difference, and it is where the exam question will be.
- **Confusing type 1 with type 2.** Type 1 replaces the OS; type 2 runs on one.

A new hypervisor host is racked. VMs in VLAN 10 work; VMs in VLANs 20 and 30 are dead.

The server team reports that of twelve VMs on the new host, the four in VLAN 10 have full connectivity and the eight in VLANs 20 and 30 cannot reach anything, including their own default gateways. The vSwitch configuration is identical to the three existing hosts, which all work. Later in the morning the port err-disables entirely and all twelve VMs drop.

A new hypervisor host is racked. VMs in VLAN 10 work; VMs in VLANs 20 and 30 are dead.

What would you check first — and what do you expect to see?

show interfaces GigabitEthernet1/0/24 switchport

SW1#

```
Name: Gi1/0/24
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Access Mode VLAN: 10 (STAFF)
Trunking Native Mode VLAN: 1 (default)
```

show interfaces status err-disabled

```
SW1#

Port       Name               Status       Reason               Err-disabled Vlans
Gi1/0/24   ESXi host 4        err-disabled psecure-violation
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. Two faults on one port, and the first explains the pattern while the second explains the timing.

The port is an access port in VLAN 10. A hypervisor uplink must be a trunk. VMs in the port group tagged VLAN 10 happen to work because the vSwitch sends their frames untagged and the access port puts them in VLAN 10 — the right answer by accident. Frames from VLANs 20 and 30 arrive tagged, and an access port discards tagged frames for other VLANs. Eight VMs are stranded.

Why it is doing that

The port also has port security. A hypervisor presents one MAC per vNIC plus the host's own management address — thirteen or more on this host. As the remaining VMs powered on through the morning, the maximum was exceeded and the port err-disabled, which is why the working four then failed too.

The tell that this is a switch-port fault and not a hypervisor fault is that the vSwitch configuration matches three working hosts. When one member of an identical set behaves differently, look at what is *not* identical — which here is the physical port it was patched into.

Correcting the uplink

IOS · configuration mode

```
interface GigabitEthernet1/0/24
 description ESXi host 4 -- vmnic0
 no switchport port-security
 no switchport access vlan 10
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 switchport nonegotiate
 spanning-tree portfast trunk
 spanning-tree bpduguard enable
!
 shutdown
 no shutdown
```

And the lesson

Fix.

The `shutdown` then `no shutdown` is required: removing the port security configuration does not clear an existing `err-disabled` state.

The process lesson. This port was almost certainly configured from the site's standard access-port template, which is correct for a desk and wrong for a server. Sites that hit this repeatedly keep a separate documented template for hypervisor uplinks — trunk, explicit allowed list, no port security, `portfast trunk`, BPDU guard — and name it in the change request.

Connecting a hypervisor properly

Platform note. You do not need a data centre. Any type 2 hypervisor on your own machine demonstrates every concept here; the switch-side tasks can be done in Packet Tracer or CML against a simulated host.

1. Identify the hypervisor you already use for the labs in this book and classify it as type 1 or type 2. Justify the classification in one sentence.
2. Create two VMs. Record the MAC address of each vNIC, and confirm they differ from the host’s physical NIC.
3. Put both VMs on a *host-only* or internal network. Confirm they can reach each other and that this traffic never appears on your physical network — capture on the physical interface to prove it. This is east–west traffic.
4. Switch both to bridged networking. Confirm they now appear on the physical network with their own MAC addresses, and find them in your switch’s MAC address table.

Connecting a hypervisor properly (continued)

1. **Switch side.** Configure the port your host connects to as an access port in one VLAN. Put one VM in that VLAN and one in another. Reproduce the Break & Fix symptom, then convert the port to a trunk.
2. Apply `switchport port-security maximum 1` to that trunk. Power on the second VM and observe. Recover the port.
3. **Containers.** Install a container engine and run two containers. Compare their start time and image size against your VMs' boot time and disk size. Record actual numbers.
4. Inspect the container network. Find the bridge, the addresses assigned, and confirm that outbound traffic is being NATted by the host. Relate this to Chapter 22.

Connecting a hypervisor properly (continued)

1. Run `uname -r` inside two containers and on the host. Explain the result, and state what it implies for isolation.
2. **Extension.** Write the switch-port template your site should use for hypervisor uplinks, with a one-line justification per command, and say what is different from your standard access-port template.

CHECKPOINT 1 of 6

What is the single structural difference between a type 1 and a type 2 hypervisor?

Discuss · then advance

ANSWER

Checkpoint 1 of 6

What is the single structural difference between a type 1 and a type 2 hypervisor?

> One layer. A type 1 hypervisor runs directly on the hardware and *is* the operating system; a type 2 runs as an application on top of a host operating system.

CHECKPOINT 2 of 6

What does a container share with its host that a virtual machine does not, and what are the two main consequences?

Discuss · then advance

ANSWER

Checkpoint 2 of 6

What does a container share with its host that a virtual machine does not, and what are the two main consequences?

The **kernel**. Consequences: containers are far smaller and start in milliseconds because there is no operating system to boot; and isolation is weaker, because a kernel vulnerability is shared by every container on the host.

CHECKPOINT 3 of 6

Why must a hypervisor uplink be a trunk?

Discuss · then advance

ANSWER

Checkpoint 3 of 6

Why must a hypervisor uplink be a trunk?

> Because the VMs on it live in several VLANs, and their frames arrive at the physical switch tagged. An access port would carry one VLAN and discard the rest, stranding every VM outside it.

CHECKPOINT 4 of 6

Why is port security inappropriate on that port?

Discuss · then advance

ANSWER

Checkpoint 4 of 6

Why is port security inappropriate on that port?

Because a hypervisor presents many MAC addresses on one port — one per vNIC plus the host's own — and they change as machines start, stop and migrate. Port security will exceed its maximum and err-disable the port, taking down every VM on the host.

What is east–west traffic, and why can you not capture it on the physical switch?

Discuss · then advance

ANSWER

Checkpoint 5 of 6

What is east–west traffic, and why can you not capture it on the physical switch?

Traffic between two VMs on the same host and VLAN. It is switched inside the hypervisor's virtual switch and never leaves the server, so it never reaches the physical switch to be seen, filtered or captured there.

A MAC address moves between two switch ports with no cable change. Give a legitimate explanation.

Discuss · then advance

ANSWER

Checkpoint 6 of 6

A MAC address moves between two switch ports with no cable change. Give a legitimate explanation.

> Live migration. A running VM was moved to another host, so its MAC address legitimately appears on a different switch port seconds later. It looks exactly like a loop and is not.

What to take away

- A hypervisor presents virtual hardware to several guest operating systems. Type 1 runs on the bare metal; type 2 runs on a host OS.
- A VM is files: a configuration and virtual disks, containing a complete guest OS with its own kernel. It can be copied, snapshotted and migrated while running.
- Each vNIC has its own MAC and attaches to a virtual switch, whose port groups carry VLAN IDs that must match the physical trunk.
- Containers share the host kernel: megabytes rather than gigabytes, milliseconds rather than seconds, weaker isolation. Orchestrators such as Kubernetes schedule them at scale.
- The two are commonly combined — containers inside VMs.

What to take away (continued)

- The physical port to a hypervisor is a **trunk** with an explicit allowed list, **no port security**, `portfast trunk` and BPDU guard.
- Live migration moves MAC addresses between ports legitimately. East–west traffic never reaches the physical network at all.

Where we got to

- A hypervisor presents virtual hardware to several guest operating systems. Type 1 runs on the bare metal; type 2 runs on a host OS.
- A VM is files: a configuration and virtual disks, containing a complete guest OS with its own kernel. It can be copied, snapshotted and migrated while running.
- Each vNIC has its own MAC and attaches to a virtual switch, whose port groups carry VLAN IDs that must match the physical trunk.
- Containers share the host kernel: megabytes rather than gigabytes, milliseconds rather than seconds, weaker isolation. Orchestrators such as Kubernetes schedule them at scale.

NEXT

Chapter 30 — Management Approaches, SNMP and Syslog